

SC2104/CE3002 Sensors & Interfacing

Tutorial 4 – DC motor and Encoder

1. A DC motor is fitted with two hall effect sensors based rotary encoder that utilizes a 24 poles magnetic disc (i.e., 12 North poles and 12 South poles) with quadrature output.
 - a) Determine the resolution (in terms of angles) if we use
 - i. X1 encoding
 - ii. X2 encoding
 - iii. X4 encoding
 - b) Two such DC gear motors with 30:1 gear ratio are used to drive a two-wheel robotic vehicle.
 - i. If the diameter of each wheel is 6cm, determine the distance travelled by the robot when the count value is equal to 1800 with an X2 encoding.
 - ii. What is the number of count value needed to make the robot travel a 1m distance?
 - iii. Comment on the resolution achieved in this system.